

Computer Vision Practical Report:

SHARPENING OF IMAGE USING UNSHARP MASKING

1. AIM

To sharpen an image using the Unsharp Masking technique in OpenCV by subtracting a blurred version of the image from the original to enhance edges, display the results, and save the output image.

2. PROCEDURE

- ☐ Load the image using `cv2.imread()`.
- ☐ Apply Gaussian blur to the image using `cv2.GaussianBlur()` to create a low-pass filtered (blurred) version.
- ☐ Subtract the blurred image from the original image to obtain the high-pass mask (edges).
- ☐ Add the high-pass mask back to the original image (or use `cv2.addWeighted()`) to produce the sharpened output image.
- ☐ Display the original and sharpened images side-by-side using Matplotlib, and save the result using `cv2.imwrite()`.

3. PROGRAM CODE

```
import cv2
import numpy as np

# 1. Load the image
image = cv2.imread('E23-IMG.png')

# 2. Blur the image to get f_bar(x, y)
blurred = cv2.GaussianBlur(image, (9, 9), 10.0)

# 3. Create the mask (Detail/High-pass filter output)
```

```

# f_s(x, y) = f(x, y) - f_bar(x, y)
mask = cv2.subtract(image, blurred)

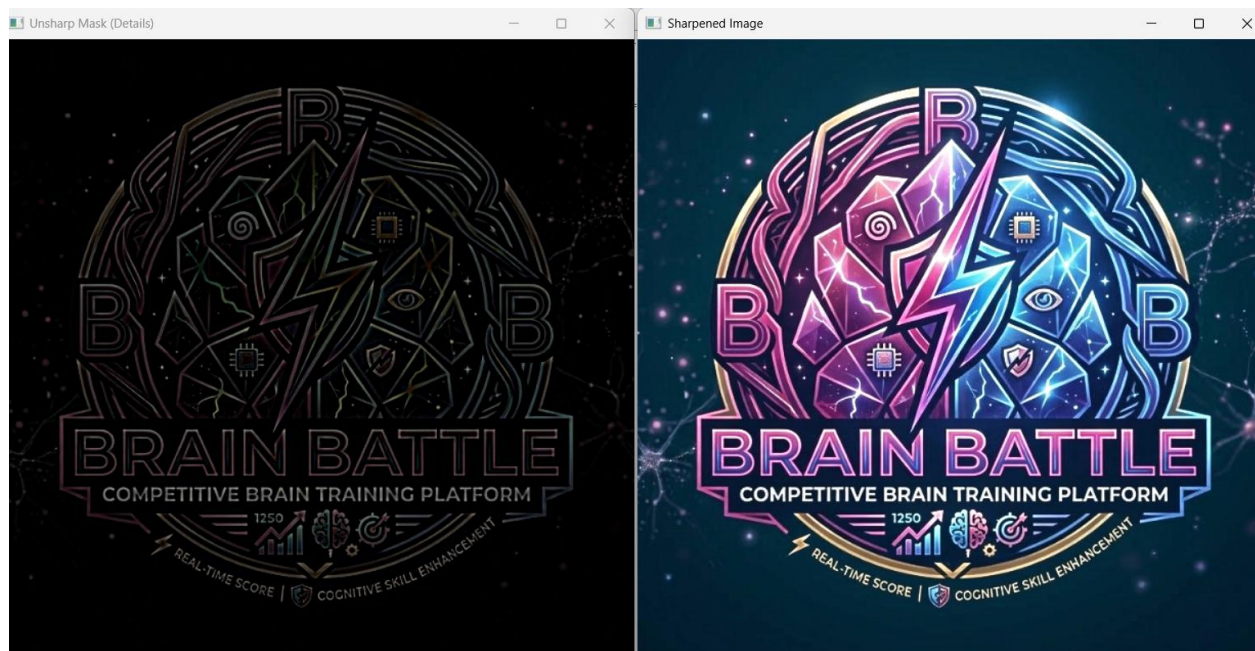
# 4. Add the mask back to the original image to get the final sharpened output
# sharpened = original + k * mask (using k = 1.0)
sharpened = cv2.add(image, mask)

# Display the results
cv2.imshow('Original Image', image)
cv2.imshow('Unsharp Mask (Details)', mask)
cv2.imshow('Sharpened Image', sharpened)

cv2.waitKey(0)
cv2.destroyAllWindows()

```

4. OUTPUT



5. RESULT

The Python script executed successfully. The input image was loaded.